

Psychometric Platform — System Architecture Overview

A Top-Down Guide to System Components, Request Lifecycles, Data Governance, and Phased Implementation

Core Stack: Spring Boot, PostgreSQL, Redis, PDF Engine

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Audience: Technical Leads, Architects, & Project Owners

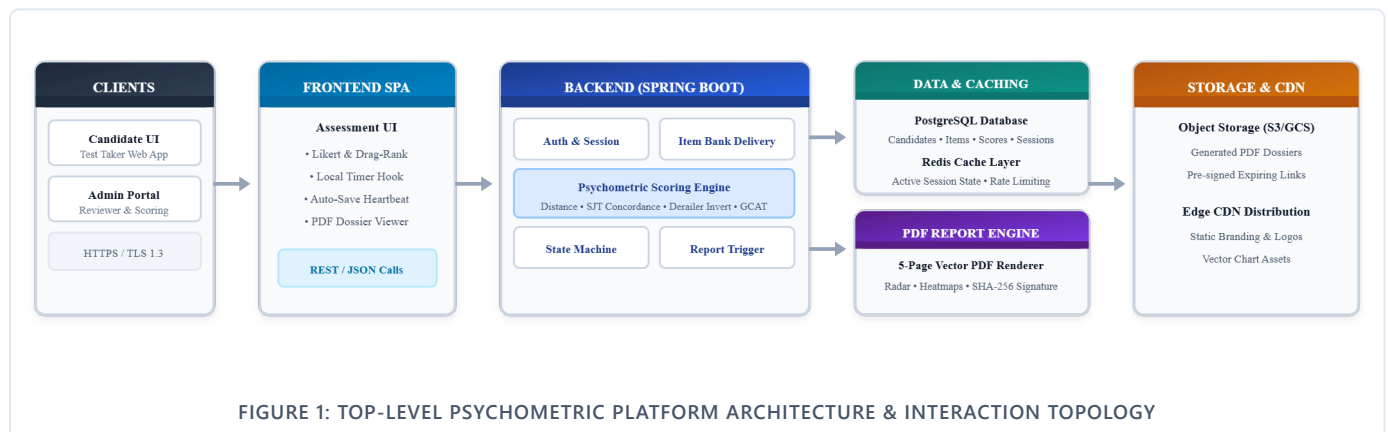
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1 High-Level System Architecture

The Psychometric Assessment Platform is designed as a secure, high-integrity assessment and reporting system. It supports candidate test administration (with strict time enforcement and anti-cheating mechanisms), item-level psychometric scoring, normative distribution scaling, and high-resolution 5-page executive PDF generation.

The system follows a modern decoupled architecture where candidate and reviewer clients communicate via a RESTful API with a containerized **Spring Boot** backend, backed by **PostgreSQL** for persistent relational storage, **Redis** for state caching, and an asynchronous **PDF Report Engine**.



Architectural Rationale: Decoupled Computation & Rendering

By isolating the Spring Boot API, caching layer, and PDF rendering engine, the platform guarantees that computationally heavy report generation will never degrade live test-taking responsiveness for candidates under timed conditions.

2 Frontend Layer & Client Experience

The client tier comprises two dedicated single-page application (SPA) interfaces tailored for distinct personas: the **Candidate Assessment Interface** and the **Admin & Reviewer Dashboard**.

A. Candidate Test-Taking UI

Adaptive Item Presenter: Dynamically renders 3 question paradigms:

- *5-Point Likert Matrix:* For PQ10 Personality and Derailer batteries.
- *Drag-and-Drop SJT Ranker:* Interactive ranking of 4 response options from Rank 1 (Most Effective) to Rank 4 (Least Effective).
- *Timed Cognitive Test Runner:* Multiple-choice speed battery.

State Resilience: Auto-saves every response in the background to prevent data loss on network disconnects.

B. Admin & Reviewer Portal

Executive Candidate Management: Provisioning assessment links, monitoring live candidate session progression, and invalidating compromised tests.

- *Interactive Dossier Viewer:* Web preview of vector radar charts, risk heatmaps, and STEN score distributions.
- *Bulk PDF Export:* Direct download of cryptographically signed PDF dossiers.

RBAC Guard: Strict permission partitioning between HR admins and reviewers.

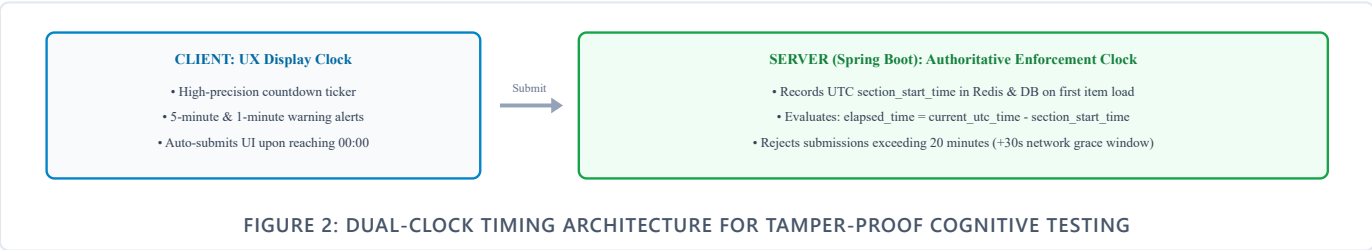
Client-Server Communication & Session Auth

The Frontend communicates exclusively with the Spring Boot backend over secure HTTPS using standard REST endpoints. Candidate test sessions are authenticated using a **single-use, time-bound Session Token** (or cryptographically signed magic link) delivered via invitation. Reviewers and system administrators authenticate via standard OAuth2 / JWT bearer tokens.

Timed Section Management: Dual-Clock Architecture (Server-Authoritative Timers)

All 4 assessment batteries are strictly timed on the server using Redis-backed timers (PQ10: 40 min / 2400s, SJT: 45 min / 2700s, Derailers: 20 min / 1200s, GCAT: 20 min / 1200s strict). Client elapsed time is never trusted. The backend initializes `battery_session:{id}:timer` upon session start. A scheduled daemon (`SessionTimeoutJob`) sweeps active sessions every 15 seconds, auto-submitting and advancing any session exceeding its cutoff limit.

A critical architectural challenge in high-stakes testing is enforcing strict time limits (e.g., GCAT's 20-minute timer). Relying solely on client-side JavaScript timers introduces vulnerability to client clock manipulation, pause-script injection, or browser sleep manipulation.



Architectural Rule: Authoritative Server Verification

The client-side clock exists solely for user feedback and smooth UX. The backend is the sole authority on elapsed time. Even if a candidate disables JavaScript or manipulates local system time, the server will reject any response submitted after `start_time + 20 minutes + grace_period`.

3 Backend Layer (Spring Boot)

The Spring Boot application serves as the central orchestration engine. It manages candidate authentication, serves test items, tracks progress via a strict state machine, executes the multi-trait scoring algorithms, and triggers PDF report generation.

Core Backend Modules

1. Auth & Session Management Module

Validates candidate access codes, creates time-bound test sessions, and enforces one-session-per-candidate isolation.

2. Item Bank Delivery Module

Serves questions for Full (258 items) or Quick (78 items) modes. Strips ideal target keys and scoring metadata from client payloads to prevent inspection tampering.

3. Assessment State Machine Module

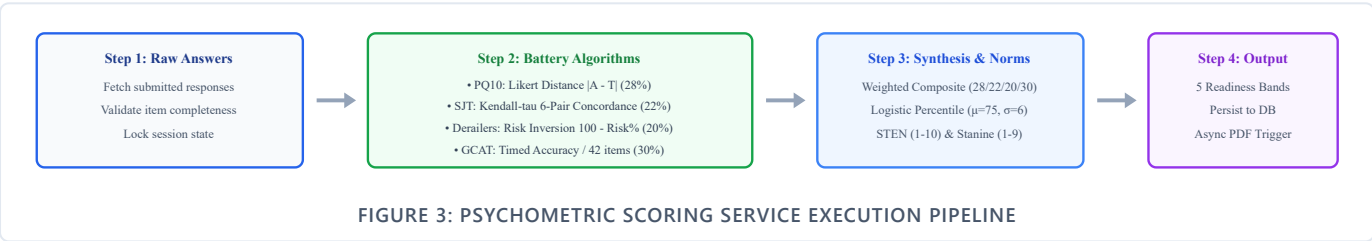
Controls session progression through immutable states (INITIALIZED → IN_PROGRESS → SUBMITTED → SCORED → REPORT_GENERATED).

4. Report Dispatcher Module

Asynchronously triggers the PDF generation engine upon test submission, transmits calculated trait STENs and composite scores, and records the resulting PDF URI and SHA-256 hash.

Placement of Scoring Algorithms in the Request Lifecycle

The psychometric scoring engine executes immediately when a candidate submits the final test section or when the session transitions to SUBMITTED. Scoring is executed in an isolated, transactional service pipeline:



Atomic Transactional Boundary

Scoring calculations and normative standardizations execute inside a single Spring transaction boundary. This ensures that scores, percentiles, and readiness bands are computed deterministically before triggering the PDF report generator.

4 Database & Persistence Layer

The persistence tier is built on a relational **PostgreSQL** database, optimized for transactional integrity (ACID), structured psychometric schemas, and audit compliance.

Core Entities & Data Sensitivity Classification

Entity Domain	What Is Persisted	Sensitivity Classification	Access & Security Governance
Candidates	Name, email, candidate ID, target role, department, metadata	Confidential (PII)	Encrypted at rest (AES-256); strict GDPR / privacy compliance.
Item Bank	Questions, battery tags, competency mapping, ideal keys (T_i)	Restricted (IP)	Internal proprietary IP; answer keys hidden from public API.
Test Sessions	Session token, exam mode (Full/Quick), start/end times, state	Internal Operational	State machine enforced; time-locked after submission.
Raw Responses	Candidate selections (A_i), scenario rankings, cognitive choices	Confidential Evaluative	Immutable record; used for psychometric DIF audit & validity checks.
Computed Scores	4 section %, STEN scores, composite %, percentile, band	Confidential Evaluative	Restricted to authorized evaluation committees and hiring managers.
Report Metadata	Generated PDF URI, SHA-256 verification hash, timestamp	Audit & Integrity	Publicly verifiable signature; tamper-evident audit trail.

5 Caching Layer (Redis)

A high-performance in-memory **Redis** instance provides fast sub-millisecond data retrieval, session state resiliency, and API traffic protection.

Essential for Launch (MVP)

- **Session Timers:** `battery_session:{id}:timer` storing server UTC start millisecond timestamp for authoritative countdown.
- **Auto-Save Heartbeat Buffer:** `battery_session:{id}:responses` caching in-flight candidate responses before batch DB flush on submit.
- **Auth OTP & Password Reset:** Temporary Redis tokens for self-registration and credential recovery with automated TTL expiration.
- **Active Session State Store:** Auto-saves every candidate answer in real time. If a candidate experiences a browser crash or network interruption, they resume immediately from their last question without data loss.
- **Server-Side GCAT Countdown:** Stores the authoritative section start timestamp in Redis with a TTL key to enforce cognitive time limits cleanly.
- **Submission Lock (Idempotency):** Prevents duplicate session submissions when clicking "Submit" multiple times.

Nice-to-Have for Scale (Post-Launch)

- **Item Bank Read Cache:** Caching the 258 psychometric questions in Redis memory to eliminate recurring DB read queries during massive concurrent testing events.
- **Rate Limiting & DDoS Defense:** Token bucket rate limiter across submission endpoints to protect against script abuse.
- **Normative Distribution Cache:** Caching percentile lookup matrices for instant percentile derivation.

Operational Benefit of Redis Session Store

Writing intermediate Likert answers to Redis takes <2ms, keeping candidate test interactions lightning-fast while buffering the primary PostgreSQL database from thousands of simultaneous write transactions.

6 PDF Report Generation Engine

The PDF Report Engine is an isolated worker component responsible for compiling calculated candidate scores into a formal, print-ready 5-Page Executive Dossier.

Architectural Workflow of Report Generation

- 1 **Asynchronous Trigger:** When a session is scored, the backend dispatches an asynchronous report generation event (via internal queue/executor), immediately returning a success status to the candidate while processing the PDF in the background.
- 2 **Data Aggregation & Template Hydration:** The engine pulls candidate demographics, 10 PQ10 STEN ratings, SJT decision metrics, derail risk heatmaps, cognitive accuracy subscales, and the promotion readiness band into an executive HTML/CSS template.
- 3 **Vector SVG Chart Rendering:** Radar charts and heatmaps are generated as vector SVG nodes to ensure crisp, zero-pixelation print clarity (300+ DPI).
- 4 **SHA-256 Signature Embedding:** A unique cryptographic hash of the candidate's scores and timestamp is computed and embedded in the report footer before finalizing the PDF.
- 5 **Storage & Secure Retrieval:** The compiled PDF is uploaded to encrypted Object Storage (e.g., AWS S3 / Cloud Storage). Reviewers download the report via time-expiring, pre-signed URLs or an authenticated backend streaming endpoint.

7 Media & Static Asset Delivery (CDN)

Unlike platforms with heavy user-uploaded images, this psychometric system primarily requires high-speed delivery of **static organizational assets**.

What Requires CDN Treatment

- **Frontend Application Bundles:** Minified JavaScript, CSS, and web fonts for the assessment SPA.
- **Report Visual Assets:** High-resolution enterprise logos, vector SVG icons, chart background watermarks, and institutional stamps used in report generation.

How the CDN Optimizes Architecture

- **Edge Caching:** Serves UI assets from geographic edge nodes close to candidates, achieving sub-second initial test load times globally.
- **Backend Offloading:** Eliminates static asset traffic from the Spring Boot API, reserving 100% of server compute for test logic and scoring.

Security Distinction: Static Assets vs. Private PDF Reports

Public CDN caching is applied **only to static branding and UI code**. Generated candidate PDF dossiers contain sensitive PII and are strictly stored in private, access-controlled buckets—never cached on public edge networks.

8 Security, Auth & Data Integrity

Executive talent assessments drive high-stakes career and organizational decisions. The system architecture embeds enterprise security, fraud prevention, and auditability at every layer:

A. Dual Persona Authentication

Candidates: Access via single-use, cryptographically secure assessment tokens tied to a specific session ID. No account registration needed; expires upon test completion.

Admins / Reviewers: Multi-factor OAuth2 / JWT authentication with strict Role-Based Access Control (RBAC).

B. Session Immutability & Anti-Cheat

State Lockout: Once a candidate clicks final submission, the session immediately transitions to SUBMITTED. Any further attempt to modify answers is rejected.

Anti-Faking Keys: Ideal answer targets (T_i) and reverse-key flags are never sent to the client browser.

C. Cryptographic Audit Signature

SHA-256 Fingerprint: A cryptographic hash combining candidate ID, raw section scores, STEN values, and timestamp is generated and stamped in the report footer.

Tamper Detection: If any score or text is altered after publication, the digital signature check fails immediately.

9 Deployment & Infrastructure Topologies

To ensure a practical implementation path, the deployment architecture is structured into two evolutionary milestones: a lightweight **Minimum to Launch (MVP)** configuration and a highly scalable **Production Scale Target**.

Infrastructure Component	Minimum to Launch (MVP Phase)	Recommended for Scale (Production Phase)
Frontend Hosting	Static SPA served via NGINX container or S3 bucket	Global Edge CDN (Cloudflare / AWS CloudFront)
Backend Application	Single Spring Boot container on standard cloud VM / App Runner	Auto-scaling container cluster (Kubernetes / AWS ECS Fargate)
Relational Database	Managed PostgreSQL (e.g. AWS RDS single-AZ, 2 vCPU, 8GB RAM)	Managed PostgreSQL Multi-AZ with Read Replicas & automated backups
Cache & Session Store	Single-node Redis container / Managed Redis instance	Redis Cluster with automatic failover and persistence (AOF/RDB)
PDF Report Renderer	Embedded headless Chrome subprocess inside backend container	Dedicated asynchronous PDF worker service on separate container queue
Report Storage	Encrypted local volume or private cloud object storage bucket	S3-compatible Object Storage with Lifecycle Rules & SSE-KMS encryption

Migration Strategy: Zero-Refactor Evolution

Because Spring Boot cleanly abstracts persistence (JPA/Hibernate), caching (Spring Data Redis), and storage interfaces, migrating from the MVP single-node deployment to the multi-container production cluster requires zero business logic rewrites—only configuration adjustments.

10 Summary: Build Priority Roadmap

To minimize project risk and establish a working end-to-end testing pipeline as quickly as possible, components should be constructed in a logical phased sequence. Below is the recommended implementation order:

Phase / Order	Component & Module	Why It Is Needed at This Step	Primary Deliverable & Output
Phase 1 (P1)	Item Bank & Data Model	Foundation for everything else. Defines the 258 questions, reverse-key flags, SJT scenarios, and cognitive items.	PostgreSQL schema, seed data scripts, and item bank delivery API.
Phase 1 (P1)	Auth & Session Manager	Enables candidates to access tests via secure single-use tokens and establishes the test session lifecycle.	Session creation endpoints, token validation, and state machine.
Phase 2 (P2)	Test-Taking UI (Frontend)	Allows actual test administration: Likert matrices, drag-and-drop SJT rankers, and cognitive timers.	Responsive candidate web app with auto-save and Dual-Clock GCAT timer.
Phase 2 (P2)	Redis Session Caching	Protects in-progress candidate answers against network loss and buffers high-frequency write operations.	Redis auto-save cache and authoritative countdown key management.
Phase 3 (P3)	Psychometric Scoring Engine	Core intelligence layer. Computes Likert distance, Kendall-tau SJT concordance, derailler inversion, and composite score.	Deterministic Spring Boot scoring service with 100% test coverage.
Phase 3 (P3)	Normative Calibration & Bands	Converts raw composite scores into Percentiles (logistic curve), STEN/Stanines, and 5 Promotion Readiness Bands.	Normative lookup service, logistic formulas, and band classifier.
Phase 4 (P4)	PDF Dossier Generator	Generates the formal 5-page executive report with vector radar charts, heatmaps, and SHA-256 digital signature.	Headless PDF rendering pipeline and encrypted report storage.
Phase 4 (P4)	Admin / Reviewer Portal	Allows HR executives and evaluation committees to review candidate results and download signed PDF dossiers.	Admin web dashboard with role-based access and batch report export.
Phase 5 (P5)	CDN & Scale Optimization	Global edge distribution for static assets, rate limiting, and multi-AZ database clustering for enterprise load.	CDN configuration, security hardening, and load test validation.

Key Milestone: Functional MVP at End of Phase 3

At the conclusion of Phase 3, the platform can administer tests, record candidate answers resiliently, compute mathematical scores accurately, and classify promotion readiness. Adding Phase 4 completes the executive PDF reporting output.

Architecture Summary & Next Steps

This architecture balances rapid time-to-market with enterprise-grade psychometric rigor, data security, and long-term scalability. The engineering team can immediately initiate **Phase 1 (Item Bank & Session Model)** knowing that all downstream dependencies, scoring mechanics, and reporting workflows have been fully mapped and architecturally validated.